

of the intended function in case of fault by implementing a combination of hardware and software solutions.

Basics of embedded systems

Fig. 1 shows the block diagram of a conventional embedded system. As you can see, all components/peripherals are tightly-coupled around a CPU into a single entity. This entity can either be a printed circuit board (PCB) or a single chip [also known as system on chip (SoC)]. Typically, FT systems have critical functional components of the system duplicated (in complex and mission-critical applications, there will be multiple units). Most commonly used mechanism is the duplication of the CPU.

Fault-tolerant system architecture characteristics

By definition, an FT system is a system that continues to function the way it is designed even in the event of component failure. The basics are:

1. Any fault that has an impact on human life or revenue is handled by FT systems. However, the level and intensity of fault-tolerance varies with criticality of applications.

For example, in a telephone exchange, when a controller fails and the system is not fault-tolerant, all subscribers connected to the exchange suffer, leading to customer dissatisfaction as well as loss of revenue due to the loss of data. So, most controllers in a telephone exchange are fault-tolerant. Whereas, if a subscriber line or the interface connecting to the exchange fails, only that subscriber is affected.

In applications in nuclear plant systems and avionics, implementation of fault-tolerance is based on the criticality of the function.

2. Fault-tolerance is implemented as a combination of hardware and software in the system.

3. Non-life-threatening FT systems are designed to handle single faults, at any given time. Technically, handling multiple faults is feasible but the cost and complexity of the

system is directly proportional to the number of faults concurrently handled. Higher the number of faults handled, more expensive and complex the system will be.

In this article we will discuss FT systems that handle only single faults. However, the philosophy of design for multiple faults is the same as single faults, barring the complexity of the design.

Understanding fault-tolerant architecture

Fault-tolerant architectures are classified into three major categories.

1. Duplication of frequently failing units (typically, power supply units)
2. Duplication of CPUs
3. CON-MON architecture (not a full-fledged FT system)

In all these systems, we need to keep in mind that the solution is a combination of hardware and software. Let us look at each of these in detail.

Duplication of power supply units (PSUs). The simplest part of FT architecture is designing a system with duplicated power supplies. This approach works in systems where power densities are high and PSUs fail frequently because of heavy load. This architecture is easy to implement as it does not call for big design changes in the main system. Fig. 2 shows the FT architecture of duplicated PSUs.

In this architecture, when one PSU fails, the other takes over. This means that, despite two PSUs being present, only one will take the full load, stressing the active PSU. This mode is also known as hot-stand-by mode.

So, to improve the architecture, modifications are

made so that PSUs share the load equally, and when one of them fails, the other takes 100 per cent of the load. This mode is known as load-sharing mode. Since PSUs in load-sharing mode are loaded only to half their capacities. This approach needs a current-sharing feature in the power supply, which makes the PSU design complex.

But there is no restriction on the number of PSUs that can be added. In telecom applications, some core systems tend to have three PSUs in the current-sharing mode so that PSUs are not loaded more than 60 per cent of their capacities to ensure reliable working.

To implement a duplicated PSU FT system, some system features need to be incorporated, such as:

1. In hot-stand-by mode or current-sharing mode, the main system should be intimated about the failure through hardware signal so that the system raises an alarm and the faulty system can be repaired.

2. When the system runs in hot-stand-by mode, the main system needs to periodically switch the PSU so that both PSUs are tested continuously.

3. In both cases, the system should have the option of PSU hot-plug-in. This ensures smooth running

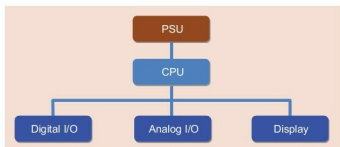


Fig. 1: Block diagram of a conventional embedded system

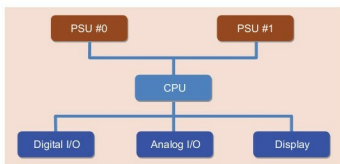


Fig. 2: FT architecture of duplicated PSUs